

Maven AI Investment Copilot – Project Overview

Problem Statement

Maven is an AI-powered investment research copilot that helps users evaluate public companies through an explainable, evidence-first pipeline instead of opaque chatbot responses.

What Maven Does

- Collects and analyzes public company financial information.
- Evaluates business quality, financial health, growth, valuation, risk, and management.
- Builds an investment thesis from deterministic analysis.
- Reviews the thesis using an AI investment committee.
- Runs an AI self-critique before the final recommendation.
- Generates both a conversational recommendation and a professional research report.
- Maintains evidence traceability throughout the pipeline.

Core Pipeline

Evidence Collection → Financial Intelligence → Investment Thesis → AI Committee Review → AI Self-Critique → Final Recommendation → Research Report

Technology Stack

- Frontend: Next.js, React, TypeScript
- Backend: Next.js API Routes
- AI Service: Python
- LLM: Google Gemini
- Architecture: Hybrid deterministic analytics + AI reasoning

Future Vision

Future versions will incorporate LangChain/LangGraph orchestration, Retrieval-Augmented Generation (RAG), vector databases for semantic retrieval, multi-agent financial reasoning, portfolio intelligence, and enterprise-grade collaboration features.